UNIVERSITY OF AGRICULTURE

Department of Computer Science

Lab 2: Programming Fundamentals (CS-308)

Basic Information			
Lab #:	2	**Teacher:**	Arsal Mahmood
Objectives	1. Single selection in C++ 2. Double selection in C++ 3. Multiple selection in C++		
Registration#		**Student Name**	
Total Marks	10	**Marks Obtained**	
Tools	Dev C++ / Visual Studio		

Activity 1 : Single Selection (if statement)

1. Write a program that takes an integer from the user and checks whether the number is even or odd using an if statement.
2. Write a program that takes a number as input and determines whether the number is positive, negative, or zero.
3. Write a program that inputs three numbers from the user and determines which number is the largest using conditional statements.
4. Write a program that takes a character from the user and checks whether it is a vowel or a consonant using logical operators.
5. Write a program that inputs a number and checks whether it is divisible by both 3 and 5 using logical operators.
6. Write a program that inputs marks of 5 subjects of a student and calculate percentage of that student. Assigns a grade based on the following conditions:

   - percentage $\geq 80 \rightarrow$ Grade A
   - percentage $\geq 70 \rightarrow$ Grade B
   - percentage $\geq 60 \rightarrow$ Grade C
   - percentage $\geq 50 \rightarrow$ Grade D
   - percentage $< 50 \rightarrow$ Fail

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7. Write a program that inputs two numbers and displays which number is greater or displays "Both numbers are equal" if they are the same.

Activity 2 : Double Selection (if-else statement, ternary operator)

8. Write a program that inputs the age of a person and determines whether the person is eligible to vote (age ≥ 18) or not eligible by using ternary operator.
9. Write a program that takes a student's marks as input and displays "Pass" if marks are greater than or equal to 50, otherwise display "Fail" (use ternary o.

Activity 3 : Multiple Selection (else ifs statement)

10. Write a program that inputs a person's age and displays whether the person is a child (age < 13), teenager (13–19), or adult (20 and above).
11. Write a program that inputs a number and checks whether the number is divisible by both 3 and 5, only by 3, only by 5, or none of them.
12. Write a program that asks the user to enter three numbers and determines whether all numbers are equal, two numbers are equal, or all numbers are different.
13. Write a program that inputs the age of a person and determines whether the person is not eligible for voting (<18), eligible (18–60), or a senior citizen (>60).
14. Write a program that asks the user to enter the marks of a student and displays Grade A (≥85), Grade B (70–84), Grade C (50–69), or Fail (<50). Use logical operators.
15. Write a program that asks the user to enter the lengths of three sides of a triangle. The program should then determine the type of triangle based on the equality of its sides.
    - If all three sides are equal, the triangle is called an Equilateral Triangle.
    - If exactly two sides are equal, the triangle is called an Isosceles Triangle.
    - If all three sides are different, the triangle is called a Scalene Triangle.

    The program should display the type of triangle according to the entered side lengths.
16. Write a program that takes a single character as input from the user. The program should determine the type of the character using conditional statements.
    - If the character is between '0' and '9', it should be classified as a Digit.
    - If the character is between 'A'–'Z' or 'a'–'z', it should be classified as an Alphabet.

- Any other character should be classified as a Special Character.

The program should display the appropriate category of the character.

17. Write a program that asks the user to enter the basic salary of an employee. The program should calculate the net salary by adding an allowance according to the following conditions:
    - If the basic salary is less than 50,000, the employee receives an allowance of 20% of the salary.
    - If the basic salary is 50,000 or greater, the employee receives an allowance of 10% of the salary.

    The program should then calculate the net salary using the formula:

    Net Salary = Basic Salary + Allowance

    Finally, the program should display the allowance amount and the net salary.
18. Develop a calculator which will calculate Sum, minus, multiplication and division of two user given numbers. It will calculate subtraction only when first number is greater than second number otherwise it should display *cannot subtract.*. Same rule is applied for division. If the second number is 0, do not divide both numbers, instead display *cannot divide a number by 0*.
19. Ask the user to enter two numbers and then ask to enter the operator (**+**, **-**, **/**, *****). Calculate and display the result (use multiple selection statements).
20. A bank gives loan in two situations
    - If customer is male and his age is more than 25 and his salary is more than 25000
    - If customer is a female and her age is more than 30 and her income is 20000

    Program: Ask user to enter age, gender(M/F) and income. Check whether he\she is eligible to get loan or not.

{end of Lab}